MIL-S-45697(Ord)
10 May 1961
SUPERSEDING
FA(FC)-PD-524
14 August 1959

MILITARY SPECIFICATION

SPOT CORRECTION UNITS, DROP-ADD: 8644080 AND LEFT-RIGHT: 8644081

1. SCOPE

1.1 This specification covers two types of spot correction units, drop-add (D-A) and left-right (L-R), subassemblies of the Gun Direction Computers: M15 and M15C, which are part of the Field Artillery Fire Control Systems: M35 and M35C. The spot correction units are used to convert mechanical inputs of observer (D-A) and (L-R) spot corrections to electrical output signals.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of
MIL-I-45208	Inspection Requirements; General Specification for

FSC 1220

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

Ordnance Corps

D8643862	Electrical Schematic, Spot Correction Unit
F8644074	Spot Correction Unit

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one each of the spot correction units, representative of production, shall be made available to the Government to determine that the spot correction units meet all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The spot correction units, drop-add and left-right shall be manufactured in accordance with Drawing F8644074, "Spot Correction Unit", and drawings pertaining thereto. The requirements of this specification are detailed only to the extent necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

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3.3.1 Ground potential.- The design and construction of the spot correction units shall insure that all external parts are at ground potential at all times.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.5 Environmental.

3.5.1 Vibration.- The spot correction units shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.03 inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the spot correction units shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.5.2 Operating temperatures.- The spot correction units shall meet the requirements of 3.6 through 3.13.3 when thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.5.3 Storage temperatures.- The spot correction units shall show no evidence of physical failure and shall meet the requirements of 3.6 through 3.13.3 at room temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 degrees and minus 65 degrees Fahrenheit.

3.6 Input knobs.- The input knobs shall rotate in clockwise directions for values of "add" and "right" and in counterclockwise directions for values of "drop" and "left" spot corrections.

3.7 Torque.- The torque required for rotation of the input knobs shall be between 10 and 20 ounce-inches. Slippage of the knobs shall occur between 2 and 3 pound-inches.

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3.8 Counters.- The spot correction counters shall be capable of rotating to at least 2495 meters in either direction from zero. One revolution of the right digit counters shall indicate a change of 100 meters.

3.9 Drum dials.- The drum dials shall indicate from 0 to 500 meters in either direction from zero.

3.10 Illumination of counters and drum dials.- When energized from a 28 volt dc source, the lamps for the windows of the counters and drum dials shall be illuminated and shall provide enough illumination to make the drum dials and counters readings legible throughout the entire range.

3.11 Backlash.- The backlash of the counter gear trains shall be no greater than 1.5 times the smallest subdivision on the counter dials.

3.12 Reset push-buttons.- The drum dials shall be capable of being reset to zero by reset push-buttons without affecting either the input sums appearing on the counters or the electrical output signals.

3.13 Electrical.

3.13.1 Circuitry.- Circuitry shall be in accordance with electrical schematic, Drawing D8643862.

3.13.2 Reference voltages.- The reference voltages supplied shall be 2.5 volts, 400 cycles per second, phased instantaneously positive and negative as referred to a common ground. They shall be applied to the upper and lower limits of the spot correction variable resistors respectively.

3.13.3 Electrical and mechanical output accuracies.- The electrical output signals shall be linearly proportional to the mechanical angular rotation of the spot correction variable resistors when loaded with 250,000 ohms, plus or minus 0.5 percent, from wipers to center taps of the variable resistors. The spot correction units shall meet the conditions and requirements specified in table I, when energized with plus or minus 2.5 volts, 400 cps.

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TABLE I

Counter Reading (meters)	Output Voltages (volts)	Tolerance (millivolts)
2490 ± .5*	2.490	± 10
2000 ± .5*	2.000	± 10
1000 ± .5*	1.000	± 10
0	0	± 10
1000 ± .5**	1.000	± 10
2000 ± .5**	2.000	± 10
2490 ± .5**	2.490	± 10

*Counterclockwise = Drop, Left
**Clockwise = Add, Right

3.13.4 Alinement and calibration.- Alinement and calibration of all spot correction units shall be determined by using the same power source to insure uniformity of product. The reference voltages supplied shall be from a power source maintained as a standard.

3.14 Identification marking.- Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's quality assurance system.- The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

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4.1.1 The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

4.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification shall consist of: (a) surveillance of the operations to determine whether the practices, methods, and procedures of the written plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviations from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrections have been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.3 Contractor's inspection system.- Contractor's inspection and Government verification of contractor's inspection shall be in accordance with Specification MIL-I-45208.

4.4 Production model.- Unless otherwise specified in the contract, one each spot correction unit shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.4.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

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4.5 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.5.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.6 Inspection of components.- All positional and functional characteristics shown on the drawings contained in the contract shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as drawing requirements affecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a precision requirement that has a bearing on the functional accuracy and performance of the spot correction units. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein.

4.6.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with table II.

TABLE II - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.6.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.7 Material.- Control of requirements of material shall be in accordance with Test Data requirements of Specification MIL-I-45208.

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4.8 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.8.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8.2 Accuracy requirements for special testing devices.

4.8.2.1 Device to test vibration requirements.- The device shall consist of a vibrating machine which will produce motion as specified. It shall be equipped with a medium capable of holding the unit securely to the vibrating machine in the specified positions.

4.8.2.2 Devices to test torque.- The device shall consist of a torque wrench type tester which will permit reading in both the clockwise and counterclockwise directions with a minimum range of from zero to twenty-five ounce-inches, the accuracy of which shall be within plus or minus one-quarter pound-inches throughout the entire zone of excursion. The device shall be equipped with an adapter to accommodate the input knob. The device for knob slippage shall be similar, but with a range of from zero to five pound-inches, the accuracies of which shall be within plus or minus one pound-inch throughout the entire zone of excursion. The device shall accommodate the same adapter.

4.9 Tests.

4.9.1 Classification of defects.- The classification of defects in tables III and IV shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

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4.9.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 4.0%		Req't Par.	QA. Par.	Test Method
101.	Post vibration	3.5.1	4.9.2	Device Special Testing (DST) Visual
102.	Operating temperatures	3.5.2	4.9.3	DST-Visual

MINOR: NONE DEFINED

TABLE IV- CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%		Req't Par.	QA. Par.	Test Method
101.	Input knobs	3.6	4.9.5	Standard Measuring Equipment (SME)
102.	Torque	3.7	4.9.6	Device, Special Testing (DST)
103.	Counters	3.8	4.9.7	Visual
104.	Drum dials	3.9	4.9.8	Visual
105.	Illumination	3.10	4.9.9	SME
106.	Backlash	3.11	4.9.10	SME
107.	Reset push-buttons	3.12	4.9.11	SME
108.	Circuitry	3.13.1	4.9.12	Visual
109.	Output accuracies	3.13.3	4.9.13	SME
110.	Identification marking	3.14	4.9.14	Visual

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4.9.2 Vibration.- This test shall be conducted in accordance with Method 201 of Standard MIL-STD-202, except that the time period for applied motion in each direction shall be as specified herein. Tests and performance measurements prior to or during vibration are not required for final acceptance. Subsequent to vibration, the spot correction units shall be subjected to a visual-tactile examination for damage. The unit shall then be subjected to all the tests in this specification.

4.9.3 Operating temperatures.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the temperature shall not be reduced beyond condition "C" or raised beyond condition "G". When the unit is stabilized at each indicated temperature, it shall be subjected to the tests specified in 4.9.5 to 4.9.13 inclusive, and shall meet the requirements specified.

4.9.4 Storage temperature.- This test shall be conducted in accordance with Procedure II, condition "B" and "H" of Specification MIL-F-13926. Upon completion of the storage temperature cycling, the unit shall be subjected to tests specified in 4.9.5 to 4.9.13 inclusive, and shall meet the requirements specified. This test shall be performed on the production model only.

4.9.5 Input knobs.- Clockwise rotation of the input knobs shall cause the counters to record increasing (add and right) values of spot corrections. Counterclockwise rotation of the input knobs shall cause the counters to record decreasing (drop and left) values of spot corrections.

4.9.6 Torque.- This test shall be performed with the aid of special testing devices, affixed to the input knob, that shall be capable of applying torque to the knob within the ranges specified. The torque required for rotation of the input knob and for clutch slippage shall be within the values specified.

4.9.7 Counters.- A visual check shall be made to determine if the specified number of mils rotation can be made.

4.9.8 Drum dials.- A visual check shall be made to assure that the dials can indicate as specified.

4.9.9 Illumination.- This test shall be performed with the lamps energized as specified. All graduations of the counters and drum dials shall be legible when observed, with the unaided eye, in total darkness.

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4.9.10 Backlash.- Energize the unit and advance the counter to some reading; record the output signal for that reading. Continue rotation of the counter in the same direction for one revolution, then reverse rotation and return until the output signal repeats the recorded reading. The counter shall have repeated its reading within the specified tolerance.

4.9.11 Reset push-buttons.- This test shall be performed concurrently with the test specified in 4.9.13. Depress the reset push-buttons after the input knobs are rotated to each rotational position listed in table I. This actuation of the reset push-buttons shall cause the drum dials to return to zero, but shall not change the input sums indicated on the counters or output voltage values.

4.9.12 Circuitry.- A visual check shall be made to assure that the circuits conform to the prescribed electrical schematics.

4.9.13 Electrical and mechanical output accuracies.- This test shall be performed with the required voltages and resistance applied as specified. Rotation of the spot correction variable resistors shall be performed with backlash excluded for direction of travel. When the counter readings of table I are obtained, the measured output voltage values shall be within the tolerance specified.

4.9.14 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The spot correction units shall be cleaned, preserved, packaged, packed and marked as follows:

(a) Spot Correction Unit, Drop-Add, 8644080-
Specification MIL-F-14619 P8535
(b) Spot Correction Unit, Left-Right, 8644081-
Specification MIL-F-14619 P8536

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6. NOTES

6.1 Intended use.- The spot correction units are functional plug-in subassemblies of the **gun direction computers(see 1.1). They** are intended to receive manual inputs of observer drop-add or left-right spot corrections and record them on total correction counters. These counters will continuously total drop-add or left-right spot corrections as they are added or subtracted in problem computation. Each change in spot correction will also appear on a drum dial so that each increment of spot correction remains recorded until changed by the computer operator. In use, the operator resets the drum dial to zero by depressing the reset button before inserting a change in spot correction. Resetting the drum dial does not affect the total correction appearing on the total correction counter; **nor** does it affect the electrical output signal.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable level of preservation, packaging and **packing**
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

6.4.1 Standard measuring equipment(SME).- Standard measuring equipment (SME) is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators,etc.

6.4.2 Device, special testing (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

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6.4.3 Tactile.- Skillful manipulation or sensing.

6.4.4 Production model.- A model in its final mechanical and electrical form of final production design made by production methods, tools, jigs and fixtures.

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